Linear & Angular Speed (4.2) Name ____________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 3 # ______ Find the arc length s for a radius $r = 5$ and central angle $\theta = \frac{\pi}{3}$.	ANSWER: 12 # ______ How many radians are in 3 complete revolutions?
ANSWER: 8π # ______ Find the arc length for $r = 6$, $\theta = \frac{\pi}{2}$.	ANSWER: 5 # ______ Convert 9 revolutions per minute to radians per minute.

ANSWER: $\dfrac{5\pi}{3}$ # ________ Convert an angular speed of 4 revolutions per second to radians per second.	ANSWER: 10 # ________ For $s = r\theta$ with $s = 20$ and $r = 4$, find θ (radians).
ANSWER: 2π # ________ Find the linear speed $v = r\omega$ for $r = 3$ and $\omega = 4$.	ANSWER: 18π # ________ For $v = r\omega$ with $v = 21$ and $\omega = 7$, find r.
ANSWER: 3π # ________ A wheel turns through 10π radians in 5 seconds. Find its angular speed (rad/sec).	ANSWER: 6π # ________ Find the linear speed of a point with $r = 2$ ft and $\omega = 5$ rad/sec.